

RG07

Agriculture and Crops of Rajasthan

Rajasthan Geography gives 11-15 questions every paper and agriculture reliably takes 2-3. The documented repeats are the National Research Centres (2016, 2018, 2023), Walra shifting cultivation (2013, 2018), hybrid varieties Mahi Kanchan and RCB 911 (2023, 2024), and crop-to-leading-district matching. Match-the-following and NOT-correctly-matched dominate.

Rajasthan farms the driest land in the country, and that single fact writes the whole chapter - coarse millets and pulses in the west, canal wheat and cotton in the north-west, soybean and oranges in the humid south-east. RPSC does not ask you about farming practice. It asks crop-to-district, crop-to-national-rank, variety-to-crop, and institution-to-city. Those four tables are the chapter.

Agriculture of Rajasthan

Seasons

- Kharif - bajra, jowar, maize, moth, guar, groundnut, cotton, til, soybean
- Rabi - wheat, barley, gram, mustard, taramira, isabgol, coriander
- Zaid - summer vegetables and fodder

First in India

- Bajra
- Rapeseed-mustard
- Guar
- Moth bean
- Isabgol
- Henna
- Cumin - largest AREA

Other ranks

- Groundnut 2nd
- Soybean 3rd
- Jowar 3rd
- Total pulses 3rd

Seed spice belt

- Cumin - Jodhpur, Barmer, Jalore, Nagaur
- Coriander - Baran, Kota, Jhalawar
- Fenugreek - Nagaur
- Isabgol - Jalore
- Ramganj Mandi - Asia's largest dhanian market

Horticulture

- Kinnow - Sriganganagar
- Orange - Jhalawar
- Mathania chilli - Jodhpur
- Sojat henna - Pali, GI tag
- Olive refinery - Bassi, Jaipur
- Date palm CoE - Sagra-Bhojka, Jaisalmer

Varieties

- Maize - Mahi Kanchan, Mahi Dhawal, Pratap Makka
- Bajra - RCB 911, RHB series, HHB 67 Improved

Research centres

- NRC Seed Spices - Tabiji, Ajmer, 2000
- NRC Camel - Bikaner
- CIAH - Bikaner
- CAZRI - Jodhpur, 1959
- CSWRI - Avikanagar, Tonk, 1962
- DRMR - Sear, Bharatpur

Universities and policy

- SKRAU Bikaner 1987 - first
- MPUAT Udaipur 1999
- SKNAU Jobner
- AU Jodhpur, AU Kota 2013
- First separate agriculture budget - Feb 2022
- 10 agro-climatic zones

Traditional systems

- Walra - Bhil shifting cultivation
- Chimata - plains and lower slopes
- Dajiya - hill slopes
- Dry-land farming, khadin

The shape of Rajasthan's agriculture - understand this and the tables make sense

Rajasthan is India's largest state by area but most of it is arid or semi-arid, and the bulk of its cropland is rain-fed. Two consequences follow, and both are examinable. First, the state specialises in crops that survive on little water - bajra, moth, guar, mustard, cumin, Isabgol, henna. That is exactly why Rajasthan tops India in those crops and not in rice or sugarcane. Second, wherever canal water arrives - Sriganganagar and Hanumangarh on the Indira Gandhi Canal - the cropping pattern flips to wheat, cotton and kinnow.

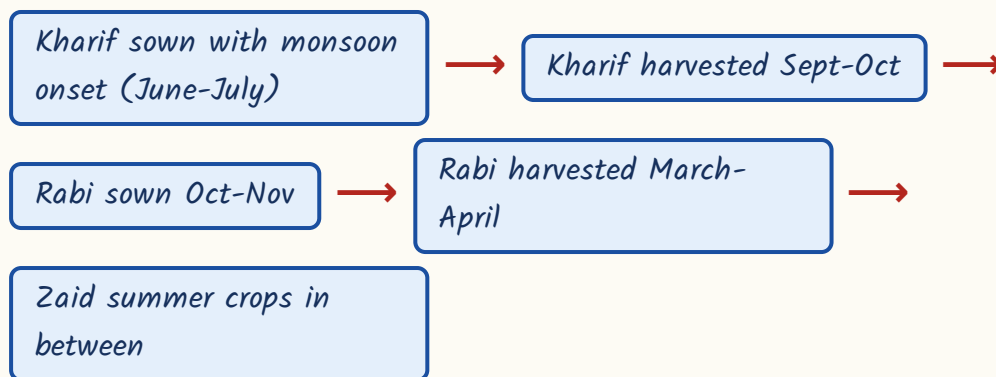
The south-east, the Hadoti plateau of Kota, Bundi, Baran and Jhalawar, is the wettest corner with black soil. That is Rajasthan's soybean, coriander and orange country. Keep this three-way division in your head - arid west, canal north-west, humid south-east - and you can reconstruct half the crop-district table without memorising it.

- Rajasthan is divided into 10 agro-climatic zones, out of 126 identified for India by ICAR.
- Zone names you should recognise - Arid Western Plain, Irrigated North Western Plain, Transitional Plain of Luni Basin, Humid South Eastern Plain.
- Kharif is the bigger season in Rajasthan by area; rabi is the bigger season by value where irrigation exists.
- Sugarcane is marginal - confined to Bundi, Sriganganagar, Chittorgarh and Udaipur; Keshoraipatan (Bundi) has the best-known cooperative sugar mill.

Kharif vs rabi in Rajasthan

Kharif (monsoon crop)	Rabi (winter crop)
Sown June-July with the monsoon, harvested Sept-Oct	Sown Oct-Nov, harvested March-April
Bajra, jowar, maize, moth, guar, groundnut, cotton, sesame, soybean	Wheat, barley, gram, mustard, taramira, Isabgol, coriander
Mostly rain-fed, arid and southern districts	Mostly irrigated, canal and well districts
Bajra is the leading crop	Mustard and wheat are the leading crops

The crop calendar - a favourite statement-based trap



TRAP

Season confusion is the standard planted error. Moth bean is KHARIF, not rabi. Barley is RABI, not kharif. Bajra is kharif. Guar and sesame are kharif; taramira and isabgol are rabi. RPSC has used exactly these swaps in 'which is NOT a rabi crop' questions.

Kharif crops - bajra is the king

Bajra is the crop of the Rajasthan monsoon. It tolerates sandy soil and low rain, so it spreads across the whole western half. Rajasthan is India's largest producer. Learn the split that RPSC loves: Alwar leads in PRODUCTION because it has better rain and yield, while Barmer has the largest AREA under the crop because it has more land and less of everything else.

Kharif crops and their leading districts

Crop	Leading district(s)	Note
Bajra	Alwar (production); Barmer (area)	Rajasthan is 1st in India
Guar (cluster bean)	Hanumangarh, then Bikaner	Rajasthan is 1st; guar gum is a major export
Moth bean	Arid western districts	Rajasthan is 1st in India
Groundnut	Bikaner	Rajasthan is 2nd in India

Crop	Leading district(s)	Note
Soybean	Jhalawar, then Baran	Rajasthan is 3rd, after MP and Maharashtra
Jowar	Ajmer	Rajasthan is 3rd in India
Maize	Southern belt - Udaipur, Banswara, Dungarpur, Chittorgarh	Tribal staple; Mahi and Pratap varieties
Cotton	Hanumangarh, Sriganganagar	Canal-irrigated north-west
Sesame (til)	Southern and south-eastern districts	Oilseed of the kharif season

- Guar gum is used as a fracturing agent by the petroleum industry - that is why guar prices swing with global oil drilling.
- Rajasthan accounts for the bulk of India's guar and guar-gum output.
- Moth, guar and bajra together define the arid-west cropping pattern.
- Rajasthan ranks 3rd in India in total pulses; Nagaur leads the state.
- Learn this four-pair set as a block - guar Hanumangarh, soybean Jhalawar, isabgol Jalore, groundnut Bikaner.

Rabi crops - mustard first, then wheat and barley

Rabi in Rajasthan means mustard above everything else. Rajasthan is India's largest producer of rapeseed-mustard, and the belt runs across the eastern plains - Sriganganagar, Alwar and Bharatpur. The research backing is examinable too: the ICAR Directorate of Rapeseed-Mustard Research sits at Sewar in Bharatpur, right inside the belt.

Rabi crops and their leading districts

Crop	Leading district(s)	Note
Rapeseed-mustard	Sriganganagar, Alwar, Bharatpur	Rajasthan is 1st in India; DRMR at Sewar, Bharatpur

Crop	Leading district(s)	Note
Wheat	Sriganganagar, Hanumangarh	Canal-irrigated north-west
Barley	Jaipur	Rabi cereal of the semi-arid plains
Gram (chana)	Bikaner	Chief rabi pulse
Taramira	Arid western districts	Hardy oilseed of poor soils
Isabgol (psyllium)	Jalore	Rajasthan is 1st in India
Coriander	Baran, Kota, Jhalawar, Bundi	Ramganj Mandi is Asia's largest market

- Mustard is sown in October-November and harvested in the spring.
- Wheat and cotton share the same home - the canal tract of Sriganganagar and Hanumangarh.
- Cotton is NOT an Udaipur crop; that wrong pair has been used as a distractor.
- Rajasthan's rabi output depends heavily on the Indira Gandhi Canal and on winter mawath rain.

Where Rajasthan ranks in India - assert only what is solid

Ranks are the most dangerous part of this chapter, because coaching material repeats old figures and actual ranks shift with the season. So separate them into two groups. Group one - the ranks that are structural and safe to state. Group two - the ranks that move, where you should recognise the crop but not bet on a number.

Solid national ranks - state these with confidence

Crop	Rajasthan's rank	Leading district in the state
Bajra	1st	Alwar (production), Barmer (area)

Crop	Rajasthan's rank	Leading district in the state
Rapeseed-mustard	1st	Sriganganagar, Alwar, Bharatpur
Guar (cluster bean)	1st	Hanumangarh
Moth bean	1st	Arid west
Isabgol	1st	Jalore
Henna (mehndi)	1st	Sojat, Pali
Cumin	1st in AREA	Jodhpur, Barmer, Jalore, Nagaur, Pali
Groundnut	2nd	Bikaner
Soybean	3rd	Jhalawar
Jowar	3rd	Ajmer
Total pulses	3rd	Nagaur

TRAP

Two rank traps. One - Rajasthan is NOT first in soybean; it is third, behind Madhya Pradesh and Maharashtra. RPSC has used 'Rajasthan is the largest producer of all EXCEPT' with soybean as the answer. Two - for cumin, Rajasthan leads in AREA; Gujarat is the other big cumin state and production leadership between the two moves year to year. Ranks for coriander, fenugreek, opium, barley and orange are quoted differently by different sources - recognise the crop, do not defend a number.

The standard framing is 'Rajasthan is the largest producer in India of all the following EXCEPT'. The answer is almost always soybean, because it sits with bajra, guar and moth bean in the option list and is the only one where Rajasthan is third.

The seed spice belt - Rajasthan's speciality

Seed spices are the crops whose seed is the product - cumin, coriander, fenugreek, fennel, ajwain, dill, isabgol. They need a dry ripening period, which is exactly what western and central Rajasthan offers. This is why the ICAR National Research Centre on Seed Spices was placed at Tabiji in Ajmer in 2000, and why RPSC keeps coming back to this belt.

Seed spices - crop to district

Spice	Belt	Marker fact
Cumin (jeera)	Jodhpur, Barmer, Jalore, Nagaur, Pali	Rajasthan has India's largest area under cumin
Coriander (dhanian)	Baran, Kota, Jhalawar, Bundi	Ramganj Mandi (Kota) - Asia's largest coriander market
Fenugreek (methi)	Nagaur	Nagaur is the fenugreek name to remember
Isabgol	Jalore	Rajasthan is India's largest producer
Henna (mehndi)	Sojat, Pali	GI tag; Rajasthan supplies almost all of India's henna

YAAD RAKHO

Yaad rakho with initials - Ja-Na-Ko-So: JAlore isabgol, NAgaur methi, KOrta-Ramganj Mandi dhanian, SOjat mehndi. Four names, four districts, and they cover most of what gets asked from the spice belt.

- Ramganj Mandi is in Kota district and is described as Asia's largest coriander market.
- Sojat mehndi carries a Geographical Indication tag.
- The NRC on Seed Spices is at Tabiji, Ajmer - established 2000, NOT at Jaipur or Jodhpur.
- Cumin's other big state is Gujarat; coriander's other big states are MP and Gujarat.

Horticulture, plantation and licensed crops

Rajasthan's horticulture is a story of matching a fruit to a micro-climate. Citrus needs irrigation, so kinnow went to the canal north-west and orange to the humid Hadoti. Arid fruits - ber, aonla, date palm - suit the west. Olive was a deliberate state experiment, and it produced a fact RPSC likes: India's first olive oil refinery.

Horticultural and commercial products to places

Product	Place	Note
Kinnow	Sriganganagar, Hanumangarh	Canal-irrigated citrus
Orange (santra)	Jhalawar	Rajasthan's orange belt in Hadoti
Red chilli	Mathania, Jodhpur	The chilli of laal maas
Henna	Sojat, Pali	GI tag
Olive and olive oil	Bassi, Jaipur	India's first olive oil refinery; Rajasthan Olive Cultivation Ltd
Date palm	Sagra-Bhojka, Jaisalmer	Centre of Excellence for date palm
Ber and aonla	Arid western districts	Backed by CIAH, Bikaner
Opium poppy	Chittorgarh, Pratapgarh, Bhilwara, Jhalawar	Licensed; regulated by the Central Bureau of Narcotics

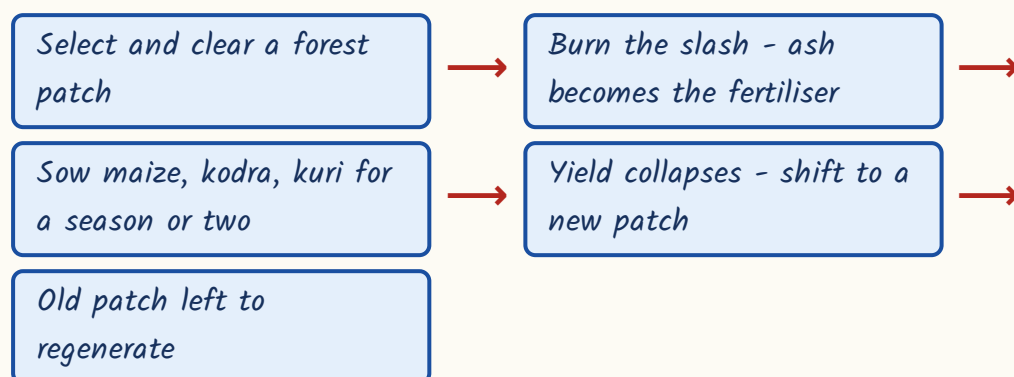
- Opium in Rajasthan is grown only under licence; the adjoining belt in Madhya Pradesh is Mandsaur-Neemuch.
- Olive cultivation is promoted through Rajasthan Olive Cultivation Ltd, a state venture.
- The Central Institute for Arid Horticulture at Bikaner works on ber, aonla, date palm and pomegranate.
- Sugarcane survives only in Bundi, Sriganganagar, Chittorgarh and Udaipur.

The four pairs that make a clean matching question out of this section are Ramganj Mandi (Kota) with coriander, Bassi (Jaipur) with the first olive oil refinery, Chittorgarh with licensed opium, and Sagra-Bhojka (Jaisalmer) with the date palm Centre of Excellence.

Walra, dry-land farming and the traditional systems

Walra is the one traditional practice RPSC has asked twice, in 2013 and 2018. It is the slash-and-burn shifting cultivation of the Bhils of southern Rajasthan - Udaipur, Dungarpur, Banswara and Sirohi. A forest patch is cleared and burnt, the ash fertilises the soil, a crop or two is taken, and then the family shifts. It has two named forms and both are asked.

The Walra cycle



The two forms of Walra

Chimata	Dajiya
Practised on plains and lower slopes	Practised on hill slopes
Larger, flatter clearings	Steeper, smaller patches
Both are forms of the same Bhil slash-and-burn system	Both are found in Udaipur, Dungarpur, Banswara, Sirohi

ASKED IN RAS

Asked in RAS Pre 2013 and again in 2018 - 'Walra' refers to shifting cultivation practised by the Bhils, and its two forms are Chimata (plains and lower slopes) and Dajiya (hill slopes). Two papers, one fact. Learn it word for word.

- Dry-land farming is the norm across western Rajasthan - moisture conservation, drought-hardy varieties, wide spacing.
- Khadin, the runoff-farming system of Jaisalmer, uses an earthen bund to trap runoff and grow a rabi crop on stored soil moisture.
- Contract farming and organic farming are promoted through state policy - the Rajasthan Organic Farming Policy came in 2017.
- Sprinkler and drip irrigation carry heavy state subsidy because of the groundwater situation.

Hybrid varieties and the National Research Centres - a

documented repeat

This section has earned marks in 2016, 2018, 2023 and 2024. Two small tables - variety-to-crop, and institution-to-city - and you can bank them. The variety question is always a NOT-correctly-matched: they attach a maize variety to bajra, or vice versa.

Hybrid and improved varieties

Crop	Varieties	Marker
Maize	Mahi Kanchan, Mahi Dhawal, Pratap Makka series	'Mahi' and 'Pratap' = maize, southern Rajasthan
Bajra	RCB 911, RHB series (Rajasthan Hybrid Bajra), HHB 67 Improved	Letter codes = bajra

ASKED IN RAS

Asked in RAS Pre 2023 - 'Mahi Kanchan' is an improved variety of maize.
 Asked in RAS Pre 2024 - 'RCB 911' is a hybrid of bajra, and the NOT-matched version planted 'Mahi Kanchan - Bajra' as the wrong pair. Two consecutive papers on the same two names.

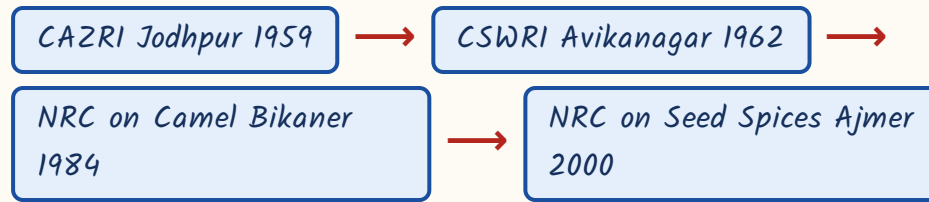
YAAD RAKHO

Trick to remember: names go with MAIZE, codes go with BAJRA. Mahi Kanchan, Mahi Dhawal, Pratap Makka - all maize. RCB 911, RHB, HHB 67 - all bajra. If the option carries a number, think bajra.

National research centres and institutes in Rajasthan

Institute	Place	Year / focus
ICAR-National Research Centre on Seed Spices	Tabiji, Ajmer	2000; cumin, coriander, fenugreek, fennel
ICAR-National Research Centre on Camel	Bikaner	1984
ICAR-Central Institute for Arid Horticulture (CIAH)	Bikaner	Ber, aonla, date palm, pomegranate
Central Arid Zone Research Institute (CAZRI)	Jodhpur	1959; desertification, arid farming, dune stabilisation
Arid Forest Research Institute (AFRI)	Jodhpur	Arid forestry
Central Sheep and Wool Research Institute (CSWRI)	Avikanagar, Malpura, Tonk	1962
ICAR-Directorate of Rapeseed-Mustard Research (DRMR)	Sewar, Bharatpur	Mustard belt

Chronology of the institutes - asked as an ordering question



ASKED IN RAS

Asked in RAS Pre 2023 - the National Research Centre on Seed Spices is at Ajmer. Asked in RAS Pre 2016 - NRC on Camel is at Bikaner, and the NOT-matched version planted 'Seed Spices - Jaipur'. Asked in RAS Pre 2018 - the Central Institute for Arid Horticulture is at Bikaner.

TRAP

Bikaner holds THREE of these - NRC on Camel, CIAH and the veterinary university RAJUVAS. Jodhpur holds CAZRI and AFRI. Ajmer holds only Seed Spices. The planted errors are 'Seed Spices - Jaipur' and 'CAZRI - Bikaner'.

Universities, policy, marketing and the current file

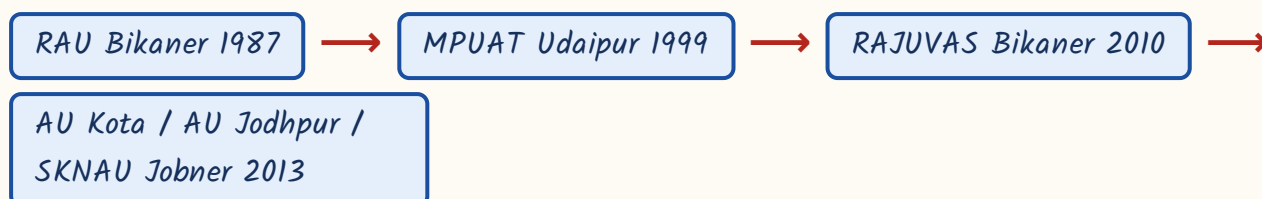
Finally the administrative layer. Rajasthan splits agricultural education by region, so each university has a headquarters town that is directly askable, plus a founding year that shows up in ordering questions. Note that the veterinary university is separate from the agriculture universities.

Agricultural universities of Rajasthan

University	Place	Year
Rajasthan Agricultural University, now SKRAU	Bikaner	1987 - the state's first
Maharana Pratap University of Agriculture and Technology (MPUAT)	Udaipur	1999

University	Place	Year
Rajasthan University of Veterinary and Animal Sciences (RAJUVAS)	Bikaner	2010
Sri Karan Narendra Agriculture University (SKNAU)	Jobner, Jaipur district	2013
Agriculture University	Jodhpur	2013
Agriculture University	Kota	2013

University chronology - an ordering favourite



- Rajasthan became the first Indian state to present a separate agriculture budget, in February 2022.
- Raj Kisan Saathi is the state portal delivering farmer services online.
- Crop insurance runs through the Pradhan Mantri Fasal Bima Yojana (PMFBY).
- The Rajasthan State Agricultural Marketing Board, 1974, runs the krishi upaj mandis.
- GRAM - Global Rajasthan Agritech Meet - was first held at Jaipur in 2016.
- Rajasthan Organic Farming Policy was announced in 2017.

CURRENT LINK

Current link, as of September 2026: Rajasthan has continued the separate agriculture budget every year since the first one in February 2022, and the 2026-27 agriculture budget again ran as a stand-alone document with a 'Developed Rajasthan 2047' crop-target roadmap and heavy sprinkler and micro-irrigation subsidy. For the exam, the fact that carries the mark is the precedent itself - first state in India, February 2022.

ASKED IN RAS

Asked in RAS Pre 2023 as a match - Seed Spices with Ajmer, Camel with Bikaner, CAZRI with Jodhpur, Rapeseed-Mustard Directorate with Sewar-Bharatpur. This is the single most reliable four-pair set in the whole chapter.

REVISION RECAP

1. Rajasthan is 1st in India in bajra, rapeseed-mustard, guar, moth bean, isabgol and henna; 2nd in groundnut; 3rd in soybean, jowar and total pulses.
2. Bajra - Alwar leads production, Barmer has the largest area.
3. Guar - Hanumangarh then Bikaner; guar gum is a petroleum-industry fracturing agent and a top export.
4. Mustard - Sriganganagar, Alwar, Bharatpur; DRMR at Sewar, Bharatpur.
5. Soybean - Jhalawar; groundnut - Bikaner; gram - Bikaner; barley - Jaipur; jowar - Ajmer; pulses - Nagaur.
6. Wheat and cotton belong to the canal north-west: Sriganganagar and Hanumangarh. Cotton is never Udaipur.
7. Cumin - largest AREA in India, western districts; Gujarat is the rival state, so do not over-claim production.
8. Coriander - Baran, Kota, Jhalawar; Ramganj Mandi in Kota is Asia's largest dhania market.
9. Fenugreek Nagaur, isabgol Jalore, henna Sojat (Pali) with a GI tag.
10. Kinnow Sriganganagar, orange Jhalawar, Mathania red chilli Jodhpur.
11. India's first olive oil refinery - Bassi, Jaipur; date palm Centre of Excellence - Sagra-Bhojka, Jaisalmer.
12. Opium is licensed in Chittorgarh, Pratapgarh, Bhilwara and Jhalawar under the Central Bureau of Narcotics.
13. Walra = Bhil slash-and-burn of southern Rajasthan; Chimata on plains and lower slopes, Dajiya on hill slopes.
14. Maize varieties Mahi Kanchan, Mahi Dhawal, Pratap Makka; bajra hybrids RCB 911, RHB series, HHB 67 Improved.
15. NRC Seed Spices - Tabiji, Ajmer, 2000. NRC Camel and CIAH - Bikaner. CAZRI and AFRI - Jodhpur. CSWRI - Avikanagar, Tonk.
16. Institute chronology: CAZRI 1959, CSWRI 1962, NRC Camel 1984, NRC Seed Spices 2000.

17. Universities: SKRAU Bikaner 1987 first, MPUAT Udaipur 1999, RAJUVAS Bikaner 2010, SKNAU Jobner / AU Jodhpur / AU Kota 2013.
18. Rajasthan has 10 agro-climatic zones out of India's 126.
19. First state in India with a separate agriculture budget - February 2022; Raj Kisan Saathi portal; PMFBY for insurance.
20. Sugarcane only in Bundi, Sriganganagar, Chittorgarh, Udaipur; Keshoraipatan (Bundi) cooperative sugar mill.